

Unit 3.3: Smoothing Frequency Domain Filters

Ideal, Butterworth, Gaussian Low Pass Filters & Laplacian Filter

Digital Image Processing Lecture

by

Kiran Bagale

Based on: Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing"

Jan, 2026

Outline: Smoothing Frequency Domain Filters

- 1 Introduction to Frequency Domain Filters
- 2 Ideal Low-Pass Filter (ILPF)
- 3 Butterworth Low-Pass Filter (BLPF)
- 4 Gaussian Low-Pass Filter (GLPF)
- 5 Comparison of Low-Pass Filters
- 6 Implementation Considerations
- 7 Effects and Applications
- 8 Summary and Comparison

Frequency Domain Filtering: Overview

- **Core Concept:** Modify frequency components to enhance or suppress specific features

- **General Process:**

- ① Compute 2-D DFT of image
- ② Multiply by filter function $H(u, v)$
- ③ Compute inverse DFT to get filtered image

- **Mathematical Expression:**

$$G(u, v) = H(u, v) \cdot F(u, v) \quad (1)$$

where $G(u, v)$ is filtered spectrum, $F(u, v)$ is original spectrum

- **Why Low-Pass (Smoothing) Filters?**

- Reduce high-frequency noise
- Blur image to smooth transitions
- Remove sharp details while preserving overall structure

Low-Pass Filters: Purpose and Characteristics

- **Objective:** Allow low frequencies to pass, attenuate high frequencies
- **Effect on Image:**
 - Smoothing and blurring
 - Noise reduction
 - Edge softening but Loss of fine details
- **Key Parameter:** Cutoff frequency D_0
 - Determines transition between passing and attenuating frequencies
 - Controls degree of smoothing and Related to distance from origin in frequency domain
- **Three Main Types to Study:**
 - 1 Ideal Low-Pass Filter (ILPF)
 - 2 Butterworth Low-Pass Filter (BLPF)
 - 3 Gaussian Low-Pass Filter (GLPF)

Ideal Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases} \quad (2)$$

where $D(u, v)$ is distance from origin in frequency domain

- **Distance Function:**

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2} \quad (3)$$

- **Cutoff Frequency:** D_0 determines filter radius

- Sharp transition at cutoff
- Frequencies within D_0 : pass completely
- Frequencies beyond D_0 : blocked completely

ILPF: Characteristics and Properties

- **Advantages:**

- Simplest to understand and implement
- Sharp frequency cutoff
- Precise control at cutoff frequency

- **Disadvantages:**

- **Ringings Artifact:** Introduces oscillations around edges
- Causes “halo” effect at sharp transitions
- Poor in practice due to Gibbs phenomenon
- Abrupt discontinuity creates spatial domain artifacts

- **Gibbs Phenomenon:**

Sharp transition in frequency $\Rightarrow$ Ringing in spatial domain

(4)

Ideal Low-Pass Filter Response

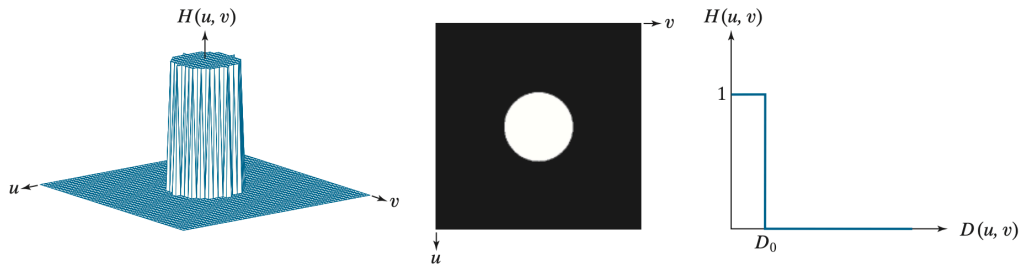


Figure: (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross section.

Gibbs Phenomenon

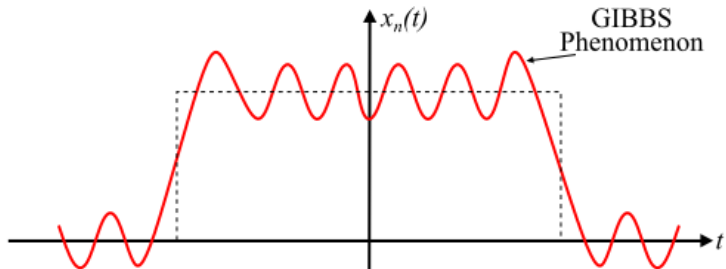


Figure: GIBBS phenomenon-clipping from the overshoots and ringing artifacts from the oscillations.

ILPF: Example Application

- **Example:** Image size 256×256 , $D_0 = 30$
- **Center at:** (128,128) (using frequency domain centering)
- **Frequency Response:**
 - All frequencies within circular region of radius 30: **pass**
 - All frequencies outside: **blocked**
- **Visual Result:**
 - Strong smoothing and blur
 - Visible ringing around edges
 - “Halo” artifacts around high-contrast areas
 - Loss of edge sharpness
- **Use Case:** Limited practical use; mainly theoretical reference

Butterworth Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}} \quad (5)$$

where $D(u, v)$ is distance and n is filter order

- **Key Parameters:**

- D_0 : Cutoff frequency (distance at which $H = 0.5$)
- n : Filter order (controls transition sharpness)

- **At Cutoff Frequency ($D = D_0$):**

$$H(D_0, v) = \frac{1}{1 + 1} = 0.5 \quad (-3 \text{ dB point}) \quad (6)$$

BLPF: Order Effect on Response

- **Order Behavior:**

- $n = 1$: Very smooth transition, gradual rolloff
- $n = 2$: Moderate transition
- $n = 4$: Sharper transition approaching ideal filter
- $n \rightarrow \infty$: Approaches ideal low-pass filter

- **Transition Characteristics:**

- No sharp cutoff like ILPF
- Smooth, monotonic decrease
- Less ringing than ILPF
- Better practical performance

- **Filter Shape:**

- Circular response in 2-D frequency domain
- Radially symmetric, i.e. Same behavior in all directions

BLPF: Visual Comparison for Different Orders

Butterworth Low-Pass Filter for Various Orders

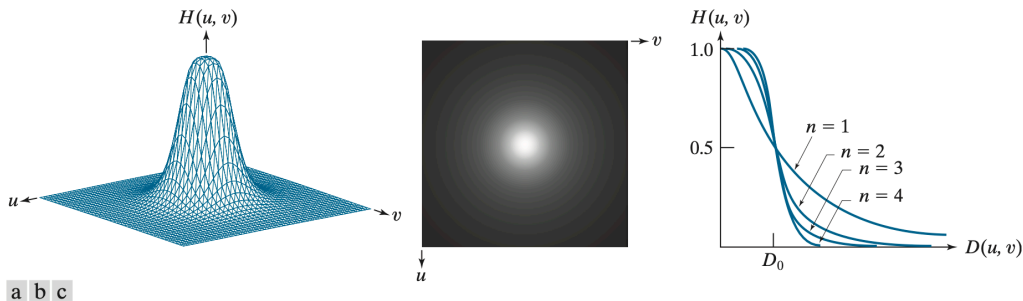


Figure: (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross sections of BLPFs of orders 1 through 4.

Higher order approaches ideal filter behavior

BLPF: Advantages and Properties

- **Advantages:**

- Smooth transition (no sharp discontinuity)
- Minimal ringing artifacts
- Adjustable transition with order parameter
- Practical and widely used
- Monotonic rolloff

- **Trade-offs:**

- Higher order $\Rightarrow$ sharper cutoff but more computation
- Lower order $\Rightarrow$ smoother but less selective
- Order $n = 2$ often good compromise

- **Typical Applications:**

- Image smoothing and noise reduction
- Preprocessing for feature extraction and General-purpose low-pass filtering

BLPF: Mathematical Properties

- **At Key Distances:**

$$D = 0 : H(0, 0) = 1 \quad (\text{DC component passes}) \quad (7)$$

$$D = D_0 : H(D_0, D_0) = 0.5 \quad (-3 \text{ dB}) \quad (8)$$

$$D = 2D_0 : H(2D_0, 2D_0) = \frac{1}{1 + 2^{2n}} \quad (\text{small value}) \quad (9)$$

- **Order Effect on Attenuation:**

$$H(2D_0, v) = \begin{cases} 0.2 & \text{if } n = 1 \\ 0.039 & \text{if } n = 2 \\ 0.0015 & \text{if } n = 4 \\ 0.00006 & \text{if } n = 8 \end{cases} \quad (10)$$

Gaussian Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = e^{-D^2(u, v)/(2D_0^2)} \quad (11)$$

where $D(u, v)$ is distance from origin, D_0 is cutoff frequency

- **Alternative Form:**

$$H(u, v) = e^{-\sigma^2 D^2(u, v)/2} \quad (12)$$

where σ controls the standard deviation (width)

- **Characteristics at Cutoff:**

$$H(D_0, D_0) = e^{-1/2} \approx 0.6065 \quad (\text{not } -3 \text{ dB like Butterworth}) \quad (13)$$

GLPF: Properties and Characteristics

- **Key Properties:**

- **Smooth:** No sharp cutoff, smooth Gaussian curve
- **No Ringing:** Spatial domain has no oscillations
- **Minimal Artifacts:** Very smooth transitions
- **Symmetry:** Both frequency and spatial domain are Gaussian

- **Relationship to Butterworth:**

- Smoother than Butterworth at all orders
- No ringing unlike Butterworth at high orders
- Easier to implement

- **Spatial Domain Equivalence:**

$$\text{Gaussian in frequency} \Leftrightarrow \text{Gaussian in spatial domain} \quad (14)$$

(Due to Fourier transform property)

GLPF: Visual Representation

Gaussian Low-Pass Filter Response

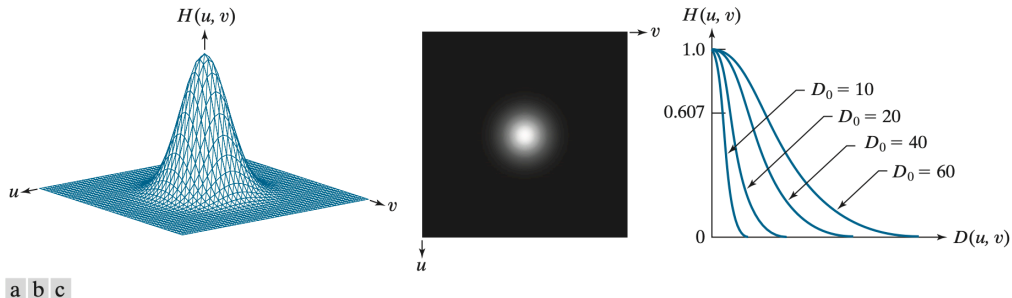


Figure: (a) Perspective plot of a GLPF transfer function. (b) Function displayed as an image. (c) Radial cross sections for various values of D_0 .

Smooth, bell-shaped response with no sharp cutoff

GLPF: Advantages and Applications

- **Advantages:**

- **No Ringing:** Smoothest possible transition
- **Best Quality:** Most natural-looking results
- **Mathematical Elegance:** Self-dual in frequency and spatial domains
- **Easy to Use:** Simple parameter (D_0)
- **Stable:** Numerically stable computation

- **Applications:**

- Image smoothing with minimal artifacts
- Pre-processing for computer vision
- Noise reduction in sensitive applications
- Medical image processing
- High-quality image enhancement

Comparison: ILPF vs BLPF vs GLPF

Aspect	ILPF	BLPF	GLPF
Formula	Sharp cutoff	$(1 + [D/D_0]^{2n})^{-1}$	$e^{-D^2/(2D_0^2)}$
Cutoff	Abrupt	At -3 dB	At $e^{-1/2}$
Transition	Discontinuous	Smooth	Smoothest
Ringing	Severe	Moderate	None
Artifacts	Halo effects	Minimal	Minimal
Complexity	Low	Medium	Low
Parameters	1 (D_0)	2 (D_0, n)	1 (D_0)
Practical Use	Theory only	Common	Preferred

Implementation: Practical Steps

- **Step 1: Prepare Image**

- Pad image to avoid circular convolution
- Typically pad to power of 2 for FFT efficiency

- **Step 2: Compute 2-D DFT**

$$F(u, v) = \text{FFT2D}(f(x, y)) \quad (15)$$

- **Step 3: Center the Spectrum**

$$F_c(u, v) = F(u, v) \cdot (-1)^{u+v} \quad (16)$$

- **Step 4: Apply Filter**

$$G_c(u, v) = H(u, v) \cdot F_c(u, v) \quad (17)$$

- **Step 5: Inverse Transform and Extract Real Part**

$$g(x, y) = \text{Real}(\text{IFFT2D}(G_c(u, v))) \quad (18)$$

Distance Calculation from Center

Standard Centering: Origin at $(0, 0)$

$$D(u, v) = \sqrt{u^2 + v^2} \quad (19)$$

Centered Coordinate System: Origin at $(M/2, N/2)$

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2} \quad (20)$$

Implementation Tips:

- Use centered coordinates for intuitive filter design
- Use `fftshift` in MATLAB/Python for centering
- Precompute distance matrix for efficiency

Computational Efficiency:

- Cache distance matrix: $D[i, j]$ pre-computed
- Avoid recomputation in filter loops
- Use vectorized operations

Choosing Filter Parameters

- **Cutoff Frequency D_0 :**

- **Small D_0** ($\approx 10 - 30$): Heavy smoothing, blur
- **Medium D_0** ($\approx 50 - 100$): Moderate smoothing
- **Large D_0** ($\approx 100 - 200$): Gentle smoothing
- **Guidelines:** Typically 5% to 10% of image dimension

- **Butterworth Order n :**

- $n = 1$: Smooth, weak filtering
- $n = 2$: Good balance (most common)
- $n = 4$: Sharper cutoff, more artifacts
- Higher n : Approaching ideal filter

- **Selection Strategy:**

- Start with Gaussian for best results
- If more control needed, use Butterworth with $n = 2$ and avoid Ideal filter in practice

Visual Effects of Low-Pass Filtering

Smoothing Effects:

- Blurs sharp transitions
- Reduces fine details
- Softens edges
- Reduces high-frequency noise

Loss of Information:

- Higher frequencies removed
- Edge sharpness reduced
- Finer structural details lost
- Trade-off with noise reduction

Ringling Artifacts:

- Most severe in Ideal filter
- Moderate in Butterworth (high order)
- Absent in Gaussian filter
- Visible as oscillations near edges

Noise Reduction:

- Remove high-frequency noise components
- Preserve low-frequency image structure
- Gaussian filter best for noise with fine structure

Image Pre-processing:

- Prepare image for edge detection
- Normalize intensity variations
- Reduce texture artifacts

Blurring Effects:

- Create soft-focus effects
- Blur background in image composition
- Artistic image processing

Medical Imaging:

- Reduce scanner noise
- Smooth speckle artifacts
- Improve image quality for analysis

Summary Table: Low-Pass Filters

Filter	Equation	Best For	Issues
Ideal	$H = \begin{cases} 1 & D \leq D_0 \\ 0 & D > D_0 \end{cases}$	Theory	Severe ringing
Butterworth	$H = \frac{1}{1 + [D/D_0]^{2n}}$	Control	Order tuning
Gaussian	$H = e^{-D^2/(2D_0^2)}$	Practice	Slower rolloff

Characteristic	ILPF	BLPF	GLPF
Ringing Artifacts	Severe	Moderate	None
Implementation	Simple	Medium	Simple
Parameters	1	2	1
Practical Use	No	Yes	Preferred
Quality	Poor	Good	Excellent